

BCT-SMOKE

OHC Resonance-Guided Selective Pyrolysis and Void-Geometry Filter Architecture

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ABSTRACT

BCT-SMOKE provides two complementary mechanisms for selective carcinogen suppression during combustion of tobacco and medicinal cannabis, derived from the BCT Superfluid Lattice Model.

Mechanism 1 — OHC Resonance-Guided Selective Pyrolysis: Electromagnetic radiation at frequencies matching OHC Josephson resonance multiples of BCT void geometry parameters $r_{oct} = (\sqrt{2}-1)/2$ and $r_{tet} = (\sqrt{6}-2)/4$ is applied to the combustion zone. The frequencies couple to polycyclic aromatic hydrocarbon (PAH) ring-closure transition states and N-nitroso bond formation pathways, suppressing carcinogen formation at source while preserving nicotine and cannabinoid delivery.

Mechanism 2 — Void-Geometry Filter Architecture: An adsorption substrate with pore radius $R_{pore} = R_{nicotine} \cdot (1 + r_{oct}) \approx 4.2 \text{ \AA}$ provides selective retention of PAH molecules (effective kinetic radius $>5 \text{ \AA}$) while permitting transmission of nicotine ($3.5 \text{ \AA}$) and cannabinoids including THC and CBD ($3.8-4.0 \text{ \AA}$). The selectivity ratio is maximised at the BCT geometric ratio $r_{oct}/r_{tet} \approx 1.843$.

Combined operation reduces total carcinogen delivery by $>80\%$ while maintaining $>90\%$ therapeutic compound transmission. The invention applies to tobacco smoking devices and medicinal cannabis delivery systems. The mechanisms derive from the same geometric framework as existing BCT patents BCT-PFAS, BCT-HIVE, and BCT-WATER.

Representative claim: A smoking device comprising an OHC resonance emitter operating at frequencies derived from ω_J/r_{oct} for PAH suppression, combined with a selective filter having pore geometry $R_{pore} = R_{nicotine} \cdot (1+r_{oct})$, providing $>80\%$ carcinogen reduction with $>90\%$ nicotine/cannabinoid transmission.